AI-Powered Smart Agro-Stake for Precision Viticulture: System Design and Implementation

Youcef HARMOUNI
Rihane DRAOU
IGEE (Ex. INELEC)
University of Boumerdes

***Abstract*—This technical documentation presents a complete engineering specification for an IoT-based agricultural monitoring system targeted at precision viticulture in Algeria. The system integrates ESP32 microcontroller architecture, environmental sensors, computer vision, and cloud-based artificial intelligence to provide real-time disease detection and environmental monitoring for grapevine cultivation. The device operates autonomously using solar power management and employs transfer learning techniques for identifying common grape diseases including powdery mildew, downy mildew, and botrytis bunch rot. System cost is optimized to $60-$80 USD per unit while maintaining commercial-grade reliability through redundant data buffering and offline operation capability.**

***Index Terms*—Precision agriculture, IoT sensors, disease detection, transfer learning, ESP32, viticulture monitoring**

I. Introduction

Algerian viticulture faces recurring challenges from fungal pathogens exacerbated by microclimate variability within vineyard canopies. Traditional monitoring approaches rely on manual inspection at intervals insufficient for early intervention. This document specifies a field-deployable monitoring system addressing three technical requirements: (1) autonomous multi-modal data acquisition from soil and canopy zones, (2) real-time image-based pathogen identification using convolutional neural networks, (3) correlation of visual symptoms with environmental conditions to establish causal relationships.

The system architecture separates edge computation (sensor reading, image capture, data transmission) from inference computation (cloud-based AI model execution) to minimize power consumption while maintaining accuracy. Data persistence at both edge and cloud layers ensures operational continuity during network interruptions.

A. Design Objectives

- Continuous monitoring: 24/7 operation under solar power
- Disease identification: Detection of three primary grape pathogens
- Cost constraint: Unit cost below $80 USD
- Environmental correlation: Simultaneous acquisition of temperature, humidity, and soil moisture
- Field durability: IP65-rated enclosures for Mediterranean climate

II. System Architecture

A. Hardware Configuration

The physical implementation consists of five integrated subsystems mounted to vineyard trellis infrastructure.

1) Control Unit:

- Microcontroller: ESP32-CAM (Espressif ESP32 SoC, 240MHz dual-core, 520KB SRAM)
- Integrated OV2640 camera module (2MP, JPEG compression)
- Local storage: 32GB microSD card (Class 10, minimum 10MB/s write speed)
- Operating voltage: 3.3V regulated

2) Sensor Array:

- Soil moisture: Capacitive sensor v1.2 (analog output, 3.3V supply)
- Air temperature/humidity: DHT22 (digital output, ± 0.5°C, ± 2% RH)
- Battery voltage monitor: Integrated ADC channel for LiPo state-of-charge estimation

3) Power Management:

- Solar panel: 6V 5W monocrystalline (156mm$\times$156mm)
- Battery: 3.7V 3000mAh Li-ion (18650 form factor)
- Charge controller: TP4056 with overcharge protection + 5V boost converter
- Deep sleep current: $<100\mu$A

4) Communication:

- Primary: IEEE 802.11 b/g/n Wi-Fi (2.4GHz, integrated in ESP32)
- Secondary: GSM/GPRS module (SIM800L, quad-band 850/900/1800/1900MHz) for remote installations

5) Physical Housing:

- Main enclosure: IP65 ABS junction box (150mm$\times$100mm$\times$70mm)
- Conduit: 20mm PVC pipe for soil sensor cable routing
- Mounting: Stainless steel clamps for trellis attachment
- Weatherproofing: Silicone sealant on all cable entry points

B. Bill of Materials

Table I provides component specifications and cost breakdown.

TABLE I
SYSTEM COMPONENT COST ANALYSIS

Component	Specification	Cost (USD)
ESP32-CAM	With OV2640 camera	8.00
MicroSD Card	32GB Class 10	7.00
DHT22 Sensor	Temperature/humidity	4.50
Soil Moisture Sensor	Capacitive v1.2	3.50
SIM800L Module	Optional GSM	8.00
Solar Panel	6V 5W	12.00
Li-ion Battery	3.7V 3000mAh	7.50
Power Management	TP4056 + boost	5.50
Housing & Hardware	Enclosure, conduit, mounts	18.00
Total		**74.00**

C. Electrical Connections

Pin assignments for ESP32-CAM module:

- GPIO 12: DHT22 data line (10kΩ pull-up resistor)
- GPIO 13: Capacitive sensor analog output via voltage divider
- GPIO 14: Battery voltage sense (1:2 voltage divider, 0-3.3V ADC range)
- GPIO 2: MicroSD card CS (chip select)
- 3V3 rail: Sensor power supply (combined draw <50mA)
- GND: Common ground for all subsystems

III. POWER BUDGET ANALYSIS

Energy consumption calculated for worst-case winter operation with reduced solar irradiance.

A. Consumption Profile

Active Mode (2 minutes per hour):

- ESP32 + Wi-Fi: 180mA @ 3.3V = 594mW
- Camera capture: 120mA @ 3.3V = 396mW (30s duration)
- Sensor reading: 35mA @ 3.3V = 115.5mW
- Peak power: 990mW
- Energy per cycle: $(594 \times 90s + 396 \times 30s + 115.5 \times 30s)/3600 = 0.24$ Wh

Deep Sleep Mode (58 minutes per hour):

- ESP32 sleep current: 0.08mA @ 3.3V = 0.264mW
- Quiescent losses: 0.5mA @ 3.7V = 1.85mW
- Total sleep power: 2.11mW
- Energy per sleep cycle: $(2.11 \times 3480)/3600/1000 = 0.002$ Wh

Daily Energy Budget:

$$E_{\text{daily}} = 24 \times (0.24 + 0.002) = 5.81 \text{ Wh}$$

B. Generation Profile

Winter Scenario (2 peak sun hours):

$$E_{\text{generated}} = 5W \times 2h \times 0.85 = 8.5 \text{ Wh}$$

Efficiency factor 0.85 accounts for panel temperature derating and MPPT losses.

Battery Reserve:

$$E_{\text{battery}} = 3.7V \times 3Ah = 11.1 \text{ Wh}$$

Provides 48-hour autonomy without sunlight: $11.1/5.81 = 1.91$ days.

IV. SOFTWARE ARCHITECTURE

A. Firmware Implementation

Firmware written in MicroPython for ESP32 platform. Core functionality implemented in single script with modular function structure.

B. Data Buffering Strategy

MicroSD card implements FIFO queue for offline data storage:

- `/buffer/pending/`: Unsynced sensor readings (JSON format)
- `/buffer/images/`: Unsynced JPEG files (filename contains timestamp)
- `/buffer/logs/`: Error logs with stack traces

On network restoration, firmware iterates through pending directory and uploads files chronologically. Successful uploads trigger file deletion to prevent storage exhaustion.

C. Device Health Telemetry

Each transmission includes metadata for remote diagnostics:

- `battery_voltage`: Current LiPo voltage (threshold alert <3.5V)
- `signal_strength`: RSSI value in dBm
- `last_sync`: Unix timestamp of previous successful upload
- `error_count`: Cumulative exceptions since last reboot
- `free_storage_mb`: Available microSD capacity

V. CLOUD INFRASTRUCTURE

A. Database Schema

Firebase Realtime Database stores structured telemetry

B. AI Model Architecture

Disease detection employs transfer learning on MobileNetV2 backbone:

1) **Base Model**: MobileNetV2 pre-trained on ImageNet (14 layers frozen)
2) **Custom Layers**: Global average pooling $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(4, Softmax)
3) **Output Classes**: Healthy, Powdery Mildew, Downy Mildew, Botrytis
4) **Training Dataset**: 3,200 labeled images from PlantVillage + custom vineyard collection
5) **Performance**: 89.3% validation accuracy, 91.2% recall on disease classes

C. Cloud Function Deployment

Firebase Cloud Function triggers on image upload

Execution time: 800ms average, scales automatically with Firebase serverless infrastructure.

VI. Field Deployment Protocol

A. Installation Procedure

1) Mount enclosure to trellis post at 1.2m height
2) Orient camera toward canopy center
3) Position DHT22 probe within leaf layer, shielded from direct sun
4) Insert soil moisture sensor 15cm depth at vine root zone
5) Route sensor cables through conduit to enclosure
6) Verify all cable glands sealed with silicone
7) Affix QR code label (URL: `dashboard.com/vine?id={device_id}`)
8) Record GPS coordinates and vineyard row/position in database

B. Device Identification System

Each unit assigned unique alphanumeric identifier (format: `VINEYARD_[ZONE]_VINE_[NUMBER]`). Physical label affixed to enclosure contains:

- Printed device ID in OCR-readable font
- QR code encoding dashboard URL with embedded device ID parameter
- Installation date and technician initials

Farmer workflow for alert response:

1) Receive push notification with device ID
2) Navigate to physical location using vineyard map in mobile app
3) Scan QR code to load device-specific data page
4) View AI prediction, sensor trends, and historical images
5) Execute intervention (fungicide application, canopy adjustment, etc.)

VII. Dashboard Interface

A. Visualization Components

Looker Studio dashboard implements four primary views:

1) **Fleet Overview**: Map showing all device locations with color-coded health status (green: operational, yellow: low battery, red: offline $>$24h)
2) **Device Detail**: Time-series graphs for temperature, humidity, soil moisture with AI alerts overlaid as event markers
3) **Disease Analytics**: Stacked bar chart showing disease detection frequency by week, filterable by vineyard zone
4) **Health Monitoring**: Table with battery voltage, signal strength, last sync time, error logs for all devices

B. Alert Configuration

Threshold-based notifications via Firebase Cloud Messaging:

- Disease detection: Immediate push notification when AI confidence $>$85%
- Battery critical: Alert when voltage $<$3.5V (approximately 20% charge)
- Device offline: Alert when no transmission received for 6 hours
- Environmental extremes: Configurable thresholds (e.g., temperature $>$38°C)

VIII. Data Security

A. Encryption

All network communication employs TLS 1.2:

- HTTPS for Wi-Fi/cellular data transmission
- Firebase SDK enforces certificate validation
- Device firmware stores server certificate hash for pinning

B. Access Control

Firebase Security Rules implement role-based authorization Device authentication via service account credentials embedded in firmware.

IX. Data Monetization Framework

A. Dual Revenue Model

1) **Hardware-as-a-Service**: Annual subscription ($150/device) includes hardware lease, cellular data plan, cloud storage, and dashboard access
2) **Data Licensing**: Anonymized dataset sales to agricultural research institutions and agritech companies

B. Anonymization Protocol

Dataset preparation pipeline for commercial licensing:

1) Remove device IDs, GPS coordinates, farmer identifiers
2) Aggregate location data to district-level granularity (minimum 10km^2 area)
3) Strip EXIF metadata from images
4) Assign pseudonymous tokens for longitudinal tracking without re-identification
5) Apply k-anonymity validation (minimum 5 farms per aggregated data point)

C. Farmer Consent

Service agreement includes explicit data usage clause:

> "Collected environmental and image data may be anonymized and included in aggregate datasets licensed to third parties for agricultural research purposes. Customer retains ownership of raw data associated with their account. Participation in data sharing program results in 15% reduction in annual subscription fee."

D. Value Proposition

Anonymized dataset applications:

- AI model training for crop disease detection
- Climate impact studies on Mediterranean viticulture
- Soil-climate correlation analysis for precision irrigation
- National agricultural statistics for policy development

X. Testing and Validation

A. Laboratory Validation

- Sensor calibration: DHT22 validated against NIST-traceable reference (Vaisala HMP110)
- Power consumption: Measured with Nordic Power Profiler Kit II
- Image quality: Assessed using spatial frequency response (SFR) targets
- AI model: 5-fold cross-validation on 800-image test set

B. Field Trial

Initial deployment: 12 units across 2 hectares (Mascara region, Algeria)

Duration: 4 months (July-October 2024)

Results:

- Uptime: 98.7% (excluding planned maintenance)
- False positive rate: 8.3% (disease alerts not confirmed by agronomist)
- Early detection advantage: 5.2 days average lead time vs. manual inspection
- Power system: Zero battery failures, minimum charge 47% during cloudy periods

XI. Future Enhancements

A. Technical Roadmap

1) **Multi-spectral imaging**: Integration of NIR camera for vegetation index calculation (NDVI, GNDVI)
2) **Edge AI**: Deployment of TensorFlow Lite model on ESP32-S3 for offline classification
3) **Mesh networking**: ESP-NOW protocol for device-to-device relay in areas with poor cellular coverage
4) **Predictive modeling**: LSTM networks for disease outbreak forecasting based on environmental trends

B. Dataset Expansion

Long-term data collection goals:

- 50,000 labeled grape disease images (5 years, 200 devices)
- Climate-disease correlation matrix for 8 Algerian viticulture regions
- Soil type classification and moisture retention profiles

XII. Conclusion

This document specifies a field-deployable IoT system for precision viticulture combining edge sensing, cloud-based AI inference, and farmer-centric visualization. The design achieves operational autonomy through solar power management while maintaining unit costs suitable for commercial deployment in developing agricultural markets. Integration of environmental sensors with computer vision enables correlation analysis between microclimate conditions and disease symptoms, providing actionable intelligence for targeted intervention.

The dual business model of hardware service and anonymized data licensing creates sustainable revenue streams while contributing to national agricultural research infrastructure. Field validation confirms system reliability under Mediterranean climate conditions with 98.7% uptime and demonstrable early detection capability.

Future work focuses on edge AI deployment for offline operation and expansion of training datasets to encompass regional cultivar variations and additional pathogen classes.

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